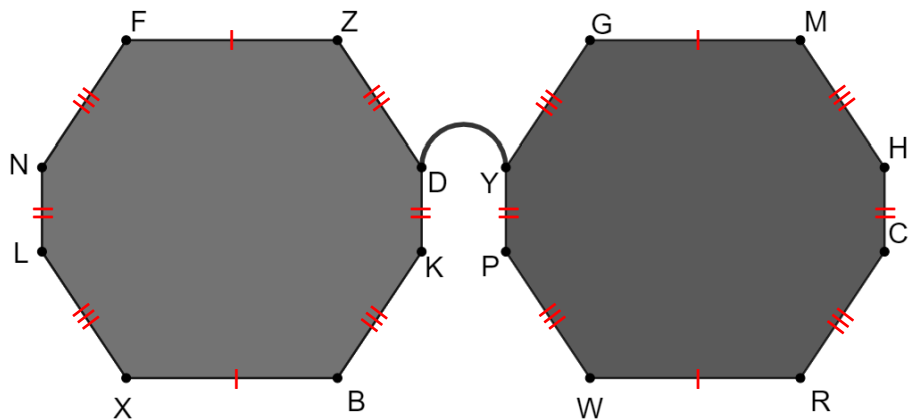


Geometry
Unit 5
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

Minnie is designing sunglasses. She uses a computer drawing program to create a diagram of the lenses. Her drawing is shown with congruent polygons $FZDKBXLN$ and $GMHCRWPY$ representing the lenses of the sunglasses.



In polygon $FZDKBXLN$, $\overline{FZ} \cong \overline{XB}$, $\overline{NL} \cong \overline{DK}$, and $\overline{DZ} \cong \overline{KB} \cong \overline{LX} \cong \overline{NF}$. The perimeter of polygon $GMHCRWPY$ is 56.8 millimeters (mm).

Some of the side measures of $GMHCRWPY$ are shown.

$$PW = 6x - 2.4$$

$$HC = 2(3x - 2.8)$$

$$WR = 5x + 2$$

$$6x - 5.6$$

- Write an equation that can be used to determine the value of x .

$$4(6x - 2.4) + 2(6x - 5.6) + 2(5x + 2) = 56.8$$

- Determine the value of x .

$$24x - 9.6 + 12x - 11.2 + 10x + 4 = 56.8$$

$$46x - 16.8 = 56.8$$

$$46x = 73.6$$

$$x = 1.6$$

- Determine the length of side GM .

$$5(1.6) + 2 = 10$$

- Determine the length of side BK .

$$6(1.6) - 2.4 = 7.2$$

- Determine the length of side NL .

$$2(3(1.6) - 2.8) = 4$$

Some of the angle measures of $FZDKBXLN$ are shown.

$$m\angle BXL = (4k - 8)^\circ$$

$$m\angle DZF = (3k + 25)^\circ$$

$$m\angle XLN = (5k - 19)^\circ$$

$$m\angle XBK = (5k - 41)^\circ$$

$$m\angle NFZ = m\angle DZF$$

$$m\angle DKB = (3k + 47)^\circ$$

$$m\angle XLN = m\angle LNF$$

$$m\angle ZDK = m\angle DZF + 22^\circ$$

6. Given that $m\angle NFZ + m\angle FZD + m\angle MHC = 394^\circ$, determine the value of k .

$$3k + 25 + 3k + 25 + 3k + 25 + 22 = 394$$

$$9k + 97 = 394$$

$$9k = 297$$

$$k = 33$$

Determine the measure of each angle.

7. $m\angle BXL = \underline{124^\circ}$
 $4k - 8$

8. $m\angle WRC = \underline{124^\circ}$
 $5k - 41$

9. $m\angle PYG = \underline{146^\circ}$
 $5k - 19$

10. $m\angle RCH + m\angle CHM = \underline{292^\circ}$
 $(3k + 47) + (3k + 47)$